

Practice Problems 6

ECON 441: Introduction to Mathematical Economics

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Exercise 5.2

1. Evaluate the following determinants:

$$(c) \begin{vmatrix} 4 & 0 & 2 \\ 6 & 0 & 3 \\ 8 & 2 & 3 \end{vmatrix}$$

$$(e) \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$$(f) \begin{vmatrix} x & 5 & 0 \\ 3 & y & 2 \\ 9 & -1 & 8 \end{vmatrix}$$

2. Determine the signs to be attached to the relevant minors in order to get the following cofactors of a determinant: $|C_{13}|$, $|C_{23}|$, $|C_{33}|$, $|C_{41}|$, and $|C_{34}|$.

3. Given $\begin{vmatrix} a & b & c \\ d & e & f \\ g & b & i \end{vmatrix}$, find the minors and cofactors of the elements a , b and f .

6. Find the minors and cofactors of the third row, given

$$A = \begin{bmatrix} 9 & 11 & 4 \\ 3 & 2 & 7 \\ 6 & 10 & 4 \end{bmatrix}$$